Aryaman Raj Jaiswal

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Education

Netaji Subhas University of Technology — Dwarka, Delhi
Bachelor of Technology in Electronics and Communications — *Aug. 2024 – Present*

Experience

SmartOut Media (Canada) — Jan. 2026 – Present
iOS Developer (Contract) — *Remote*

- Built the **SmartOut** iOS app from the ground up in **Swift/SwiftUI** (**50K+ installs**) — an offline-first geospatial platform for Ontario hunting and fishing regulations — over a bundled **SQLite** dataset and a migratable **GRDB** store, instrumented with **GA4** analytics and **Crashlytics**.
- Engineered the map layer with **Google Maps SDK** and **GEOSwift**: **456 regulatory polygons** with MULTIPOLYGON and interior-hole support, marker clustering, polygon hit-testing, and **deep links** restoring the exact zone, tab, and regulation row across launches.

Bharti Airtel Limited — June 2026 – Aug. 2026
Software Engineer Intern — *Gurugram, India*

- Re-architected an internal **OpenStack** platform tracking state across a distributed VM fleet; cut dashboard query latency by **30%** via **database migration**, **composite indexing**, and **cursor-based pagination**.
- Built a guardrailed natural-language interface over the VM database: **LangChain**, a read-only **text-to-SQL** agent, and **NeMo Guardrails** rails.

Examination Cell, NSUT — May 2025 – Aug. 2025
Software Developer Intern — *Delhi, India*

- Shipped **NSUTrack**, a native **iOS** client (plus an Android counterpart) centralizing schedules, attendance, and notices for **1,500+ students**, on an offline-first cache with **FCM** push.
- Architected the **Next.js**/**Flask** platform behind it for **10,000+ students**, cutting manual data entry **70%** with **Pandas** constraint solvers.

Projects

Wandr | *Swift, SwiftUI, Network.framework, Swift Concurrency, Foundation Models, App Intents, Google Places/Routes API* 2026

- Built the real-time multiplayer layer directly on **Network.framework** with **zero third-party dependencies** — an `NWConnection` WebSocket client against an `NWListener` relay — bridging callbacks into an **AsyncStream** consumed by a `@MainActor` `@Observable` room, with a **generation counter** retiring stale callbacks so a cancel cannot double-schedule a reconnect, **0.25s → 4s backoff**, identity replay, and an offline outbox flushed on reconnect — all on a `nonisolated`, serial-queue-confined transport under **MainActor-by-default isolation**, which those callbacks would otherwise inherit and trap on.
- Designed the relay to hold **no authority**: it rebroadcasts a versioned snapshot and every device folds it through the same **pure tally** (plurality, quorum gate, deterministic ties), so all clients converge on the identical decided schedule without exchanging a winner.
- Engineered an `actor`-isolated **seven-stage planning pipeline** where **Foundation Models** return **array indices only** into a pre-vetted deck — never names, prices, or hours — under `@Generable`/`@Guide` constrained decoding with a **12s TaskGroup timeout**; splitting one 13-field generation into three raised correctness **∼12% → ∼72%**. Venue research runs over **Google Places/Routes** behind a swappable `VenueResearching` protocol; covered by **340 Swift Testing cases**.

AquaSense | *Swift, SwiftUI, Speech, AVFoundation, FastAPI, LangChain, Gemini, Random Forest, Computer Vision* Jan. 2026

- Built the **SwiftUI** client with a fully **on-device voice assistant** — **Speech framework** transcription and **AVSpeechSynthesizer** playback — over a stateless **FastAPI** backend.
- Grounded every **Gemini** claim in **Random Forest** output to eliminate hallucinated toxicity verdicts, behind a **multi-agent LangChain** layer routing between telemetry health, **computer-vision** disease, and marketplace agents.

Achievements

- **Apple Swift Student Challenge Winner (2026)** for Saldo, an on-device personal finance app built with **no third-party dependencies** (SwiftUI, **Vision**, **VisionKit**, **Foundation Models**): **VisionKit OCR** receipt scanning and **Vision** subject masking feed a **versioned image pipeline** with automatic migration, and OCR and inference run **actor-isolated** off the UI thread with **zero cloud dependency**.
- **Winner, MLH Brainwave 2.0 (DTU):** Led Team Aquacult to **1st place over 200+ teams**, sweeping **Best Overall Project**, **Best Use of Gemini**, and **Best UI/UX Runner-Up**.

Technical Skills

Languages: Swift, Python, Kotlin, JavaScript/TypeScript, SQL (PostgreSQL), C/C++
iOS: SwiftUI, UIKit, Swift Concurrency (actors, async/await, AsyncStream, Sendable), Network.framework, App Intents, Core ML, Vision, VisionKit, Speech, AVFoundation, MapKit / Google Maps SDK, Core Data, GRDB/SQLite
Testing & Tooling: Swift Testing, XCTest, Instruments, OSLog / signposts, TestFlight & App Store release, Xcode, Git
On-Device AI: Foundation Models (@Generable / @Guide), Core ML, Vision OCR, NLContextualEmbedding
Backend: FastAPI, Flask, Node.js, REST APIs, PostgreSQL, Firebase (Auth/FCM/Analytics), Crashlytics